

Implement Security Monitoring and Incident Response

Final Report — Detection Use Case and Incident Response Scenario

Consolidated from the HVMC Network Security Rubric report and the JOMANGY tabletop IR exercise

Project Name	Healthcare IAM Operations Simulation
Organization	Horizon Valley Medical Center (HVMC)
Rubric	Implement Security Monitoring and Incident Response
Source Documents	Network_Security_Rubric_Report.pdf (\$7); HVMC_Tabletop_IR_JOMANGY_Scenario.docx
Document Version	1.0
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1. Rubric Cross-Reference

This report consolidates both required components of the “Implement Security Monitoring and Incident Response” rubric line, drawing on two already-completed HVMC lab deliverables so each scenario stands on real, previously-documented detection and response work rather than being written from scratch.

Rubric Requirement	Section	Evidence
Monitoring use case: detection rules, alert prioritization, response procedures	Section 2	Suricata brute-force detection rule + alert prioritization table
1 documented incident: classification, response steps, lessons learned	Section 3	JOMANGY tabletop IR exercise (Detect-Contain-Eradicate-Recover-Lessons Learned)

Note on addressing: both source documents were written using the lab's originally planned 192.168.50.0/24 scheme. The lab's actual working subnet was subsequently corrected to 192.168.64.0/24 (see the project's network reconciliation notes). The scenarios below are reproduced as originally documented; a full address update across all deliverables is a tracked, pending item.

2. Security Monitoring Use Case

2.1 Detection Rule

A Suricata IDS rule was deployed on the lab's virtual network to detect brute-force authentication attempts against the domain controller (HVMC-DC01):

```
alert tcp any any -> $DC01_IP any (msg:"Possible Brute Force - Repeated Auth Failures"; \
flow:to_server; threshold:type both, track by_src, count 15, seconds 10; \  sid:1000010;
rev:1;)
```

This rule alerts when more than 15 authentication-adjacent connection attempts arrive from a single source within a 10-second window — consistent with automated credential-guessing tools (e.g., Hydra, Crowbar) rather than normal human login behavior.

2.2 Alert Prioritization Process

Priority	Criteria	Example
High	Targets the domain controller or a privileged account; sustained over multiple attempts	Brute-force against a decoy privileged account on HVMC-DC01
Medium	Targets a standard user account; single-source, non-sustained	A handful of failed logins from one workstation
Low	Informational probe with no follow-through	A single port scan with no subsequent authentication attempt

2.3 Response Procedure by Priority

- High: immediate triage, contain the source (block IP / disable account access path), verify no successful authentication occurred, escalate for review within the hour.
- Medium: triage within the shift, confirm account lockout status, note in the tracking log; escalate to High if the pattern recurs or targets a privileged account.
- Low: log for trend analysis; no immediate action unless it correlates with another alert.

3. Incident Response Scenario — Hypothetical JOMANGY Compromise

Full detail lives in the companion document HVMC_Tabletop_IR_JOMANGY_Scenario.docx; this section extracts the classification and response-lifecycle content required by this rubric. The scenario is explicitly framed as a tabletop exercise — a “what would happen” walkthrough against the JOMANGY malware analyzed under the Identify and Analyze Cyber Threats rubric — not a record of an actual incident.

3.1 Incident Classification

Incident Type	Malware compromise — initial access via phishing, followed by multi-stage backdoor deployment (JOMANGY/INJ3CTOR3 family)
Entry Vector (assumed)	Phishing-based credential capture (ties to the Sep 4 SET Credential Harvester deliverable), rather than JOMANGY's real-world direct internet-facing exploit path — adapted because IAMLab-Net is outbound-only with no inbound route
Severity	High — six-layer self-healing persistence and lateral-movement risk toward the domain controller on a flat network
Affected Scope (hypothetical)	A single foothold host, with lateral-movement risk extended to HVMC-DC01 given the lab's flat, unsegmented network design

3.2 Response Steps Taken

Detect	Correlate Suricata alerts, Samba audit-log anomalies, Wireshark capture, and OpenVAS findings against a samba-tool configuration baseline diff — no new tooling required beyond what the lab already runs.
Contain	Isolate the foothold host from the network segment; block outbound C2 traffic at the firewall; preserve volatile evidence before further action.
Eradicate	Remove all six persistence layers (cron entries, shell-profile injections, immutable crontab backups, watchdog process, redundant web-shell copies, self-reinstalling PHP executor) — partial cleanup is explicitly called out as insufficient given JOMANGY's self-healing design.
Recover	Rebuild the affected host from a clean UTM snapshot rather than trusting an in-place clean; rotate any credentials that were reachable from the compromised host.
Lessons Learned	Network segmentation to prevent flat-network lateral movement; disciplined clean-snapshot practice for fast rebuild; a beacon-interval Suricata rule tuned to JOMANGY's C2 pattern; reinforced phishing-awareness tie-in

	given the assumed entry vector; regular patch cadence.
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3.3 Alert Prioritization for This Incident

Classified High per the criteria in Section 2.2: the assumed lateral-movement path threatens HVMC-DC01 directly, and JOMANGY's self-healing persistence means the threat is sustained rather than a single event.

4. Evidence of Functionality

The detection rule and alert-prioritization framework in Section 2 were implemented and demonstrated as part of the Network Security rubric submission (Network_Security_Rubric_Report.pdf, §7), including the underlying brute-force IDS alert and Samba audit-log correlation. The incident scenario in Section 3 was fully authored and delivered as HVMC_Tabletop_IR_JOMANGY_Scenario.docx. This report cross-references both so the Security Monitoring and Incident Response rubric has a single, standalone submission rather than requiring the grader to assemble it from two other documents.

5. Conclusion

Both components of the Implement Security Monitoring and Incident Response rubric are satisfied: Section 2 demonstrates a working detection rule, a documented alert-prioritization process, and priority-based response procedures; Section 3 documents a full incident scenario with classification, a five-stage response lifecycle, and concrete lessons learned.

6. References / Sources

- Network_Security_Rubric_Report.pdf, Section 7 — Monitor and Respond to Network Security Events.
- HVMC_Tabletop_IR_JOMANGY_Scenario.docx — full tabletop incident response exercise.
- Malware_Diagnosis_Report_JOMANGY_679515b3.docx — underlying malware analysis referenced by the incident scenario.